

Frequency Modulation Based on Hyperbolic Tangent Function for Position Closed Loop Control of Stepper Motor

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ABSTRACT

Closed-loop position control of stepper motors is highly non-linear. The authors propose a model-free controller utilizing a hyperbolic tangent function (Tanh) to modulate the driver input frequency and direction. Experimental results show a 60% tracking error reduction compared to traditional PI algorithms when tracking sinusoidal trajectories, providing smooth acceleration and deceleration with minimal stopping oscillations.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Provides advanced math and algorithms for the closed-loop position control of the stepper motors using magnetic encoders (AS5600) on both the ESP32 S3 and Arduino Uno nodes.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Wenlin Zhou provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.